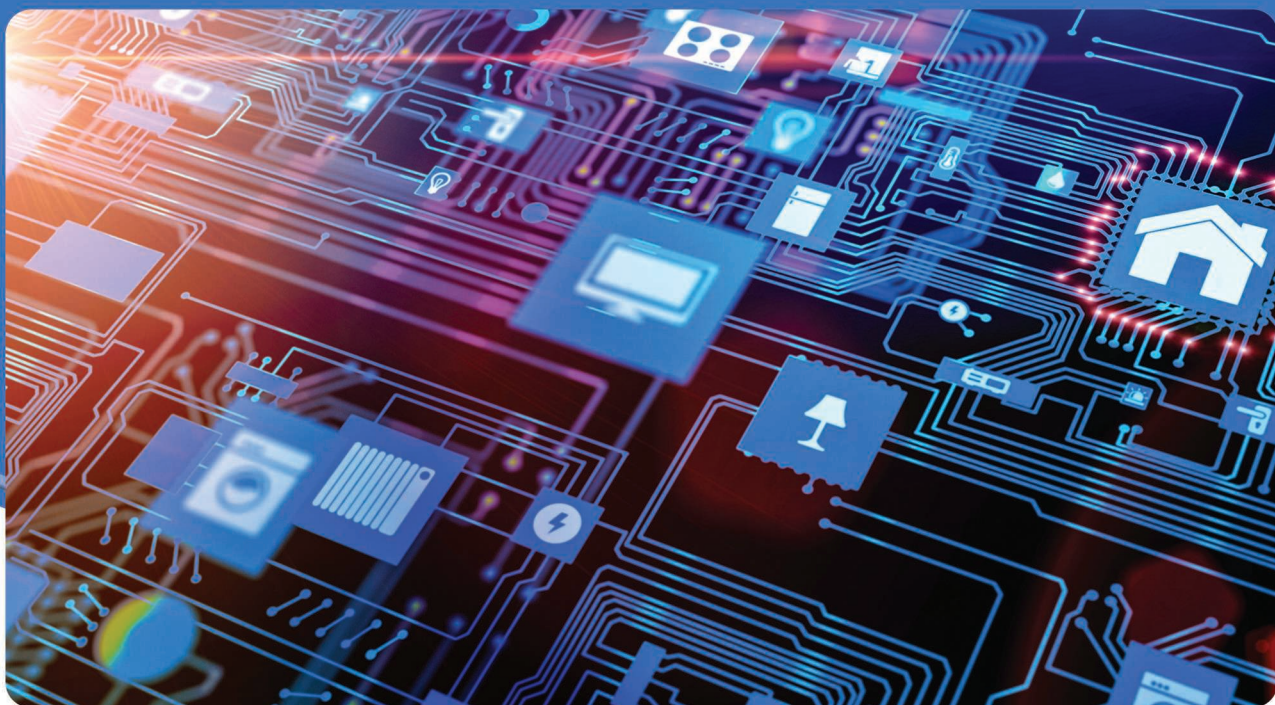




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Enabling Data-Driven Condition Monitoring of Power Electronic Systems With Artificial Intelligence: Concepts, Tools, and Developments

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Introduction

Various measures can be applied in the design, production, and operation phases of PES to fulfill reliability and availability requirements. Due to the limitations and uncertainties in models and data used in the design and production phases, the risk of unexpected failure events still exists. It calls for proactive actions that can be facilitated with condition monitoring, where the health status of PES is monitored to identify and mitigate potential risks [1]. Moreover, the obtained in-situ operation status can also be used for operation optimization of PES, if it is not necessarily for preventing unexpected failure.

Condition monitoring methods can be divided into model-based and data-driven. Model-based methods rely mainly on the deterministic aspects of power electronic components and systems, such as circuit operation models and degradation models. Besides these established models, uncertainties and unexplored system principles exist in terms of noise factors, environmental conditions, unknown degradation mechanisms, just to name a few. These unknown aspects make model-based methods challenging. Artificial Intelligence (AI) provides an opportunity to overcome such a challenge. Figure 1 shows the basic concept of AI-assisted model generation. Model-based methods use the inputs and established models to obtain the outputs. Data-driven methods can apply AI algorithms and a set of existing input data (with or without output data) to generate the hypothesized models, which would be otherwise undetermined or unexplored. Then these generated models are used to obtain the corresponding outputs for new sets of input data. For the condition monitoring of PES, the inputs could be electrical, thermal, mechanical, or acoustic signals, and the outputs could be estimated component parameters, stress

conditions, degradation level, remaining useful lifetime (RUL), normal or abnormal status, etc.

The condition monitoring in PES has been drastically evolved in the last decade with recent advancements in AI [2], [3], industrial Internet-of-Things (IIoT), and data analytics. Data-driven condition monitoring solutions have been emerging as a promising direction. They show significant potentials and merits, e.g., improved accuracy, flexibility, and lower requirements on hardware circuitry and deterministic models.

Cost, implementation complexity, accuracy, and impact on existing system operation are primary factors to benchmark condition monitoring methods. Meanwhile, it should be noted that any added hardware circuitry or software algorithm for condition monitoring implementation brings new risk to the system of concern, which needs to be minimized. Several challenges in data-driven condition monitoring for PES are discussed as below:

- *Limited size of data set:* PES are not a data-intensive area compared to other data-driven applications like image recognition or natural language processing. For example, the time-to-failure data is scarce since the device degradation is a very long process of more than several years in regular usage; accelerated testing experiment is resource-consuming (e.g., time, facilities, manpower) as well. Moreover, PES are usually in high-frequency operation; the required resolution and sampling frequency of electro-thermal stresses may not be fulfilled by the existing data set for condition monitoring purposes.
- *Heterogeneous operational environment:* PES are usually operating in a wide range of climatic conditions and dynamic loading conditions. These external factors affect the patterns of the collected data. It results in the data with high-level variability, including system-to-system variability, temporal variability, measurement errors, etc.
- *Limited computational platform:* the computational platform in the PES is traditionally designed for a control purpose, which is of limited computation power to support complex data analytics. The execution time

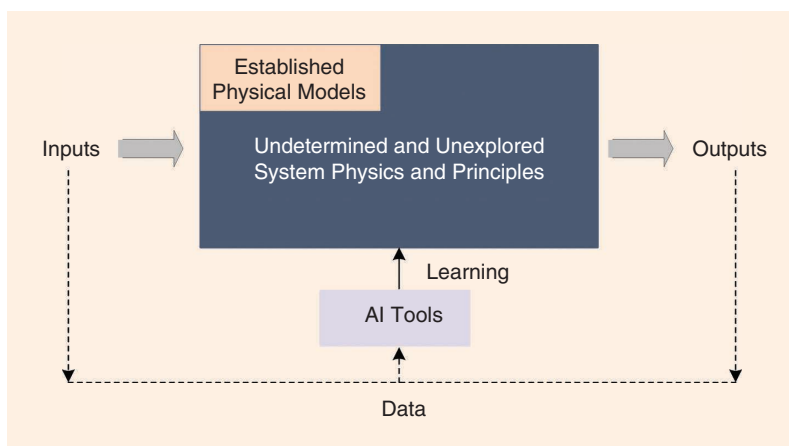


FIG 1 AI-assisted model generation.

requirements of condition monitoring on PES are diverse, covering from insensitivity to time-criticality. For example, some condition monitoring applications, e.g., fault detection, are time-critical and require real-time decision-making.

- **Mission-critical applications:** most of the condition monitoring applications on PES are mission-critical, and the tolerance for false prediction is low. As a result, it requires the explainability, trustworthiness, and credibility of the data-driven solutions, which can provide highly reliable results.

The above challenges demand holistic considerations in data, AI algorithms, and application constraints when delivering the benefits of data-driven solutions to PES.

Data-Driven Condition Monitoring Concepts and AI Tools

Electrical or electro-thermal data are usually used as the initial inputs for the condition monitoring of PES. These could be: 1) existing available data for control and protection purposes, such as converter input/output voltages and currents, dc-link voltage, heat-sink temperature, and case temperature; and 2) additional measured data dedicated to condition monitoring, such as the on-state satu-

ration voltage of an IGBT module and ripple current of a capacitor. Figure 2 illustrates the procedure to apply the data and AI for RUL prediction, anomaly detection, and degradation state identification. There are two key steps. Step 1 aims to obtain relevant health indicators or features, and Step 2 aims to perform the targeted prediction based on the health indicators or features. Current I , voltage V , and temperature T represent available initial data in the PES of interest for the illustration purpose.

In Step 1, there are two scenarios. The first one is that the initial data are directly used as the input of Step 2. The second one is often the initial data themselves cannot be directly used as health indicators or features to ensure sufficient prediction accuracy in Step 2. For the latter scenario, either data-driven or model-based methods can be applied. In [5], an Artificial Neural Network (ANN) based method is discussed to estimate the capacitance value of capacitors based on the dc component and ac component of the dc-link voltage of a three-phase motor drive. It requires training data consisting of dc and ac components of the dc-link voltages under different dc-link capacitances. To avoid such a training process, health indicators for RUL and statistical features for anomaly detection are obtained

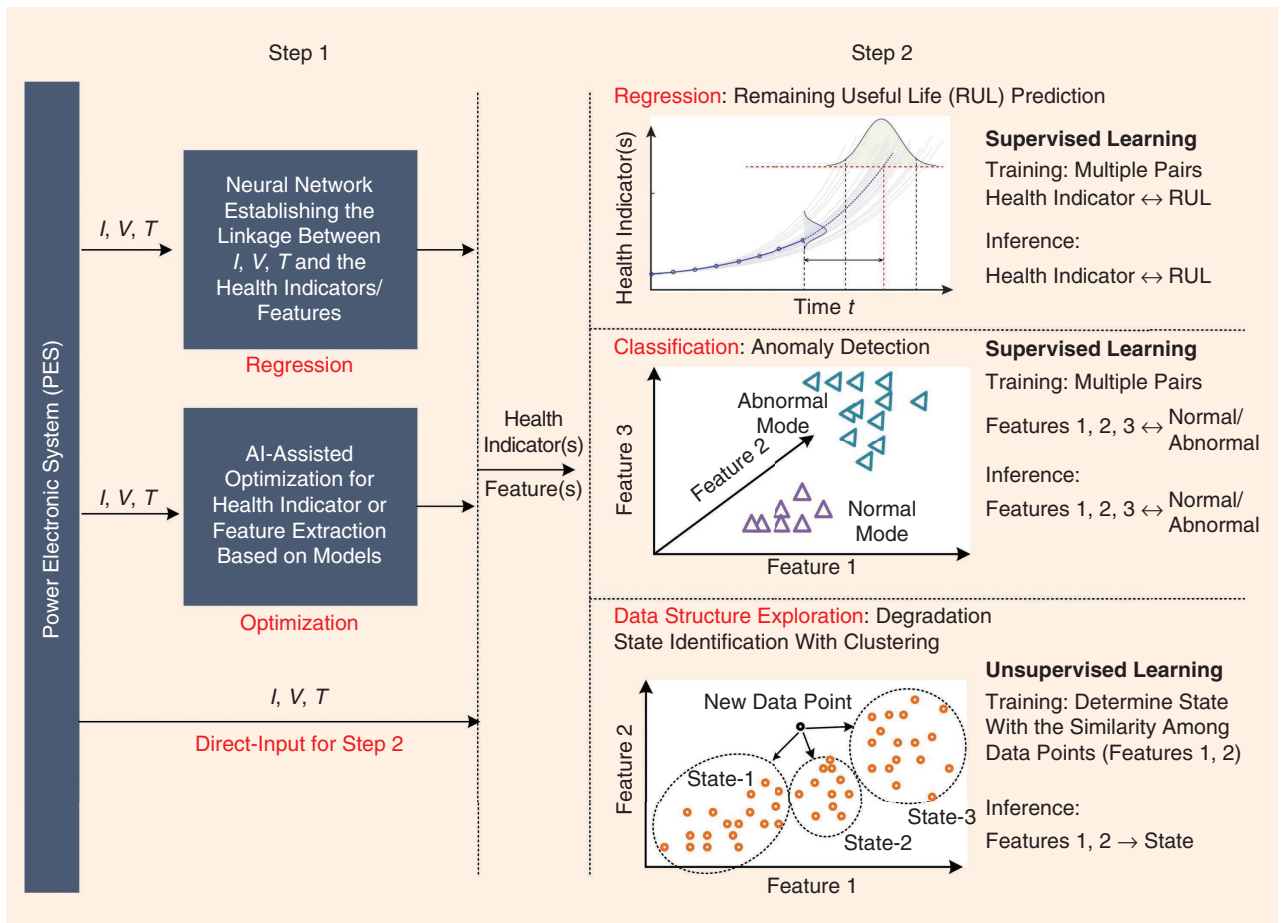


FIG 2 Illustration of different functions of AI in condition monitoring of power electronic systems.

based on models and optimization algorithms in [6] and [7], respectively. The two case studies will be introduced in Section 3 in detail.

The health indicators or features are intermediate information for different condition monitoring applications shown in the three examples in Step 2. RUL prediction is based on predicting the trends of relevant health indicator(s). Anomaly detection is based on the classification of different features. The two examples in Figure 2 apply a set of known data between the health indicator and RUL or features and the normal/abnormal status, i.e., labeled data, for the training, which is called supervised learning. The goal of it is to establish functional relationships between the input-output pairs. If there is no target output in the data set for training, it is unsupervised learning. The goals of unsupervised learning include data clustering and data compression. For example, to determine degradation states based on similarities in historical data or among different PES.

The AI applications shown in Figure 2 can be considered to solve four types of problems: optimization, regression, classification, and data structure exploration. The detailed explanations of them can refer to [4]. Various AI tools can be used for solving the respective problem. The review articles [2], [4] indicate that AI has been continuously applied in PES since the 1990s for design, control, and maintenance. Notably, the maintenance applications, relevant to condition monitoring, have an evident increase since 2015 [4]. In 2019, a tutorial on AI applications to power electronics [8] gives a comprehensive introduction of AI applications in PES. Generally, AI tools are categorized as expert system (ES), fuzzy logic (FL), metaheuristic methods, and machine learning (ML). The most active area of AI applications for PES is ML, with a percentage of 45.8%, according to an investigation in [4]. This article does not focus on giving a detailed discussion of each type of these AI methods. For details refer to [2], [4], and [9]. Instead, the following section will present case studies, which briefly introduce the applied AI tools.

State-of-the-art Developments

Case Studies

A. PSO for Optimization Problem in Estimating Health Indicators of a Buck Converter

In [6], a digital twin concept is proposed for estimating the health indicators of the MOSFET, capacitor, and inductor in a buck converter. Figure 3 shows the case study. The buck converter digital twin includes mathematical models of the power stage, feedback signal conditioning stage, and control stage. The mathematical models are system differential equations. The inductance L , the equivalent series resistance of inductor R_L , capacitance C , the equivalent series resistance of the capacitor R_C , and drain-to-source on-state resistance of the MOSFET $R_{ds(on)}$ are of interest. The available data from existing sensors are the inductor current I_L and output voltage V_o . The internal component parameters of the physical buck converter and its digital twin are considered identical if their inductor current and output voltage are the same. Thus, estimating the internal component parameters is equivalent to solving an optimization problem,

$$\underset{L, R_L, C, R_C, R_{ds(on)}}{\operatorname{argmin}} f_{\text{obj}} = \frac{\sum_{j=1}^N [(I_{L,p,j} - I_{L,d,j})^2 + (V_{o,p,j} - V_{o,d,j})^2]}{N} \quad (1)$$

where $I_{L,p,j}$ ($V_{o,p,j}$) and $I_{L,d,j}$ ($V_{o,d,j}$) are the sampled inductor current (output voltage) of the physical Buck converter and its digital twin at sampling time t_j , respectively. N is the number of sampling points. A metaheuristic method, particle swarm optimization (PSO) is applied to find the optimal solutions of (1), i.e., a set of values of L , R_L , C , R_C , and $R_{ds(on)}$, by narrowing down the difference close to zero between the measured I_L and V_o from the physical prototype and those calculated from the digital twin. As a result, that data of interest can be indirectly obtained without any additional sensor hardware.

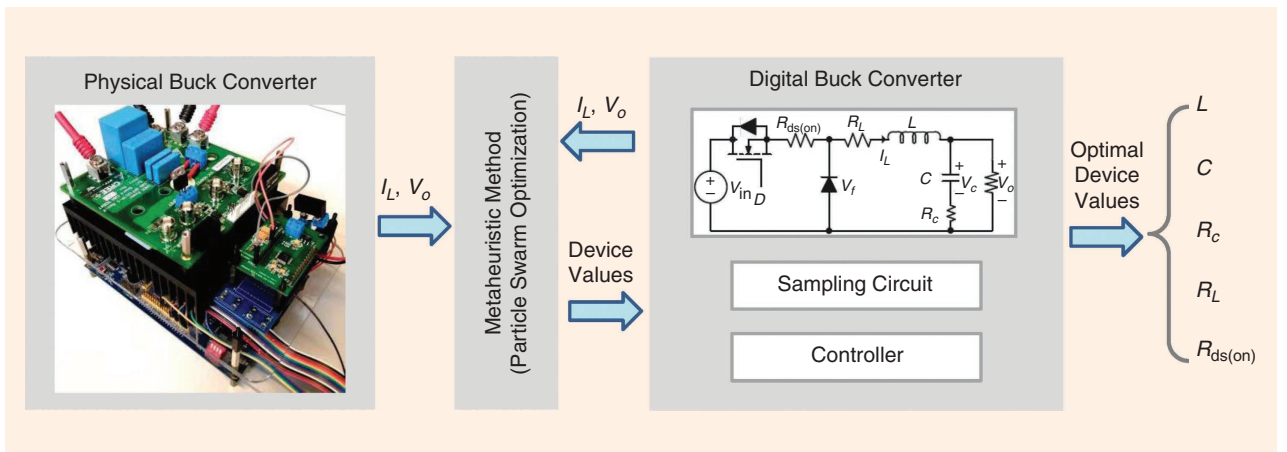


FIG 3 Digital twin-based health-indicator estimation for a Buck converter [6].

B. Composite Failure Precursor (CFP) Optimization for SiC MOSFETs Based on Information Fusion

This case study shows how to obtain a single optimal health indicator, i.e., CFP in [10], based on multiple sources of possibly relevant data. An optimal CFP expects to achieve improved accuracy and reduced uncertainty in RUL prediction based on a single source of failure precursor. Experimental analysis indicates that multiple physical parameters are related to the degradation of SiC MOSFETs [10]. In [10], a metaheuristic method, genetic programming (GP), is applied to integrate multiple physical parameters of SiC MOSFETs (i.e., the input capacitance C_{iss} , output capacitance C_{oss} , reverse transfer capacitance C_{rss} , drain-source on-state resistance $R_{ds(on)}$, gate threshold voltage $V_{GS(th)}$, and diode forward voltage V_{SD}) to formulate a CFP. As a symbolic programming method, GP can automatically search for a nonlinear combination form of the physical parameters that minimize a specific objective function, e.g., the prediction error in this case. The candidate combination form consists of explicit mathematical operations such as add, minus, times, divide, logarithm, trigonometric functions, etc. More advanced operations can be added into the combination forms to make the GP more flexible and powerful. As a result, a composite failure precursor is derived, as shown in Figure 4. Compared to the widely used parameter $R_{ds(on)}$, the prediction accuracy by using the developed composite failure precursor is improved by 35.3%, and the associated uncertainty level is reduced by 16.3%, which indicates the great potentials of information fusion. It is worth mentioning that there is no physical implication of the fusion function in Figure 4. This function is derived from a data-driven perspective.

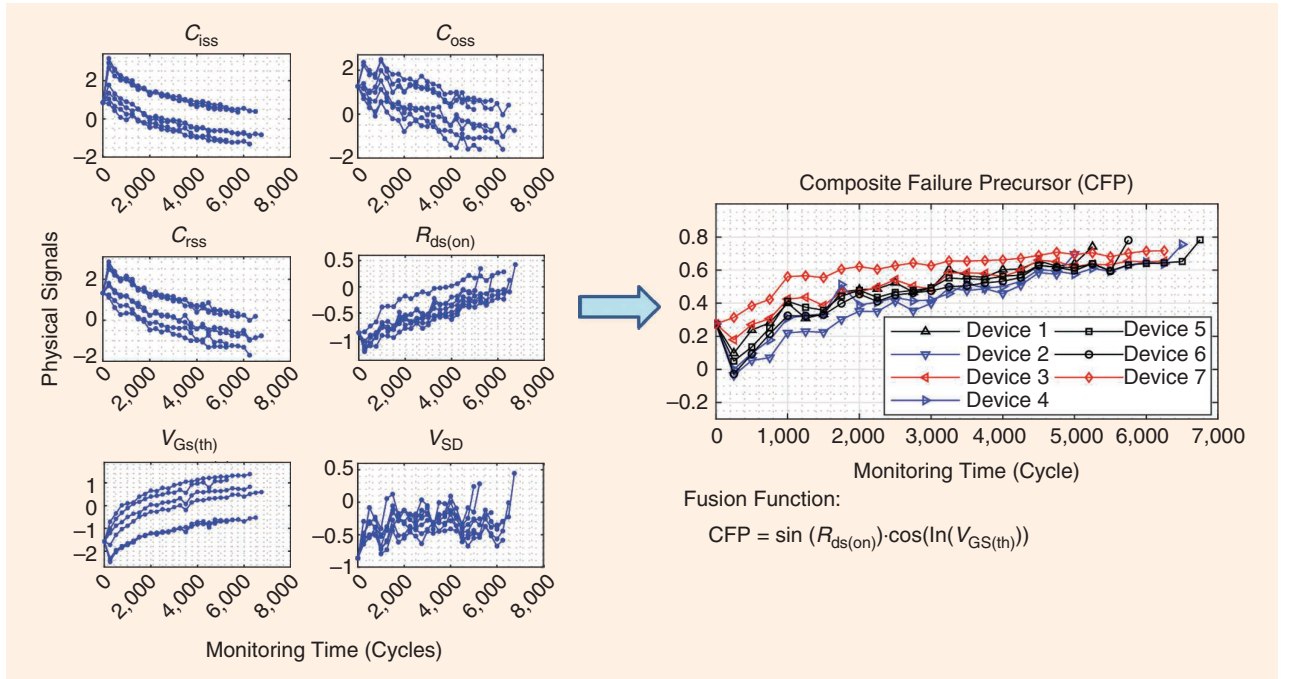


FIG 4 Composite failure precursor based on information fusion integrating multiple data sources [10].

C. Early Degradation Prediction and Testing Time Reduction of Batteries

Figure 5 shows a case study presented in [11] on the early degradation prediction and testing time reduction of batteries. AI tools are applied to determine the high-cycle-life charging protocol that maximizes Lithium-ion battery lifetime in the battery cycling experiment, as shown in Figure 5. Generally, the battery life is highly affected by the charging protocols determined by three independent parameters CC1, CC2, and CC3. These three parameters formulate a total of 224 charging protocols. It is very resource-consuming to determine the high-cycle-life protocol by extensively exploring the whole 224 candidates.

An early-prediction model enabled by a ML, elastic network, is applied to predict the lifetime with the first few cycles of data. The elastic network establishes the mapping relationship between the features extracted from the first 100 cycles and the final battery lifetime. Meanwhile, the elastic network can automatically select the relevant features to improve the generalizability and interpretability of the method. As a result, the final lifetime (average lifetime 905 cycles) of the batteries can be predicted by a model trained using the data of the first 100 cycles. It suggests that it is unnecessary to continue the experiment after only 100 cycles for estimating the lifetime of a new battery, which significantly saves the testing time. Subsequently, a metaheuristic method, Bayesian optimization, is used to optimize the design parameter space by balancing the exploration of high uncertainty area and the exploitation of the predicted high-cycle-life. With an active learning capability, Bayesian optimization can purposefully search for the input space with a high-level uncertainty where there

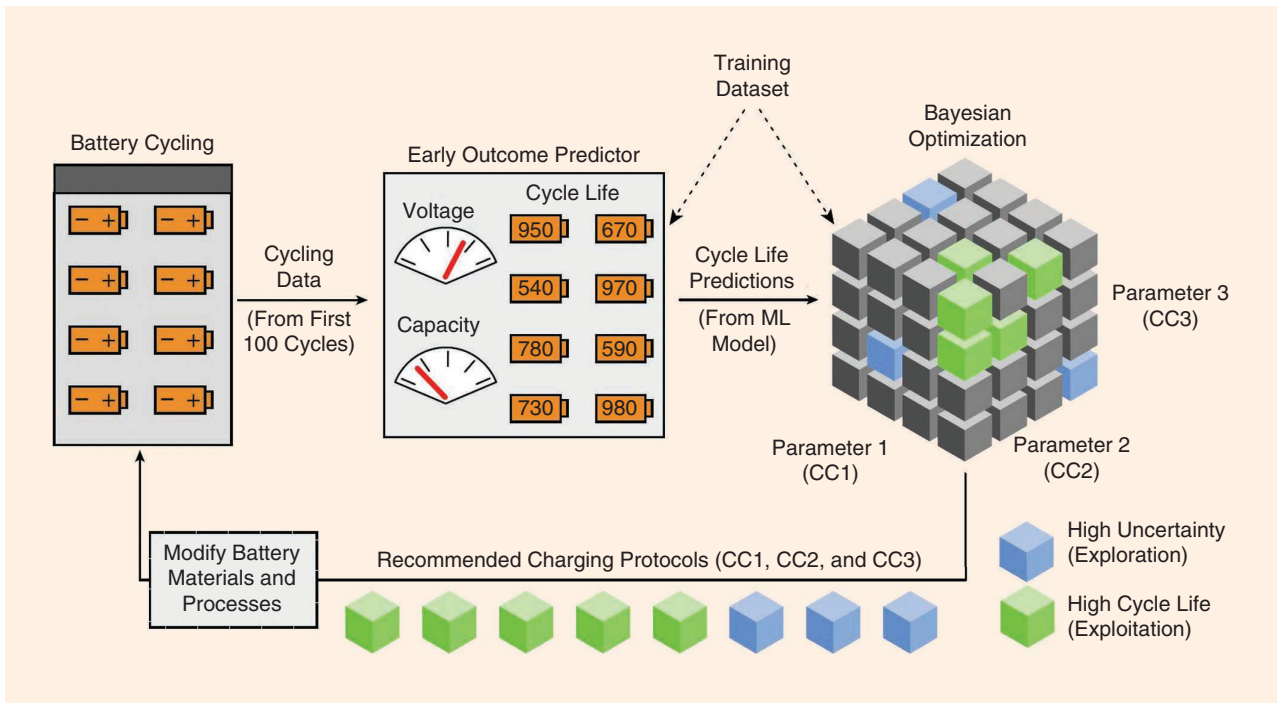


FIG 5 Procedure of the AI-assisted experiment design for time-efficient Lithium-ion battery testing [11].

is a high probability to obtain the optimal high-cycle life. Meanwhile, this high-level uncertainty can be reduced with more data from this space. With this recommendation, as a closed-loop, the subsequent experiment design can specifically focus on this input space and collect more training data to reduce the model uncertainty. In this case, the recommended charging protocols (i.e., the optimal combination of parameters 1, 2, 3 shown in Figure 5) are obtained from the optimization results, which serve as a guide to modify the experiment setting in the next experiment iteration. By avoiding uninformative testing, the experimental parameter space can be optimized to reduce the number of experiments. As a result, the high-cycle-life charging protocols are rapidly identified from 224 candidates in 16 days, compared to more than 500 days required by a conventional testing method. Even though this case study primarily focuses on testing, the early degradation prediction method could be extended to component degradation prediction in the operation phase. It also demonstrates a time-efficient testing method for collecting degradation data, which could be used for condition monitoring.

D. Online Anomaly Detection in DC/DC Converters Based on Multiple Statistical Features

In [12], a case study on anomaly detection in dc-dc converters is presented. Seven statistical features are obtained in Step 1 shown in Figure 2, from the output voltage of dc-dc converters with unknown topology. They are the range, mean, standard deviation, skewness, kurtosis, entropy, and the centroid of the output voltage, as defined in [12]. Gaussian process regression (GPR) method is applied to learn the normal behavior of a dc-dc converter for anomaly detection.

It can provide probabilistic outputs with inherent uncertainty quantification capability, as shown in Figure 6. First, the previous 300 sampling data points of the output voltage are applied for the model training, resulting in a model indicating the normal behavior of the converter. This model can generate the normal voltage output, as the data points between 300 and 500. The GPR also provides the 95% confidence interval of the normal voltage to indicate the normal operational range, i.e., the uncertainty quantification capability. The extreme values of the seven statistical features are used to determine the anomaly behavior. Due to the

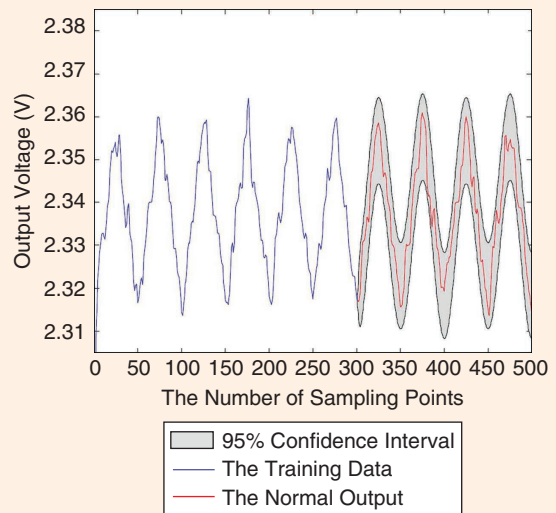


FIG 6 Estimated output voltage of a dc-dc converter based on the Gaussian process regression (GPR) model [12].

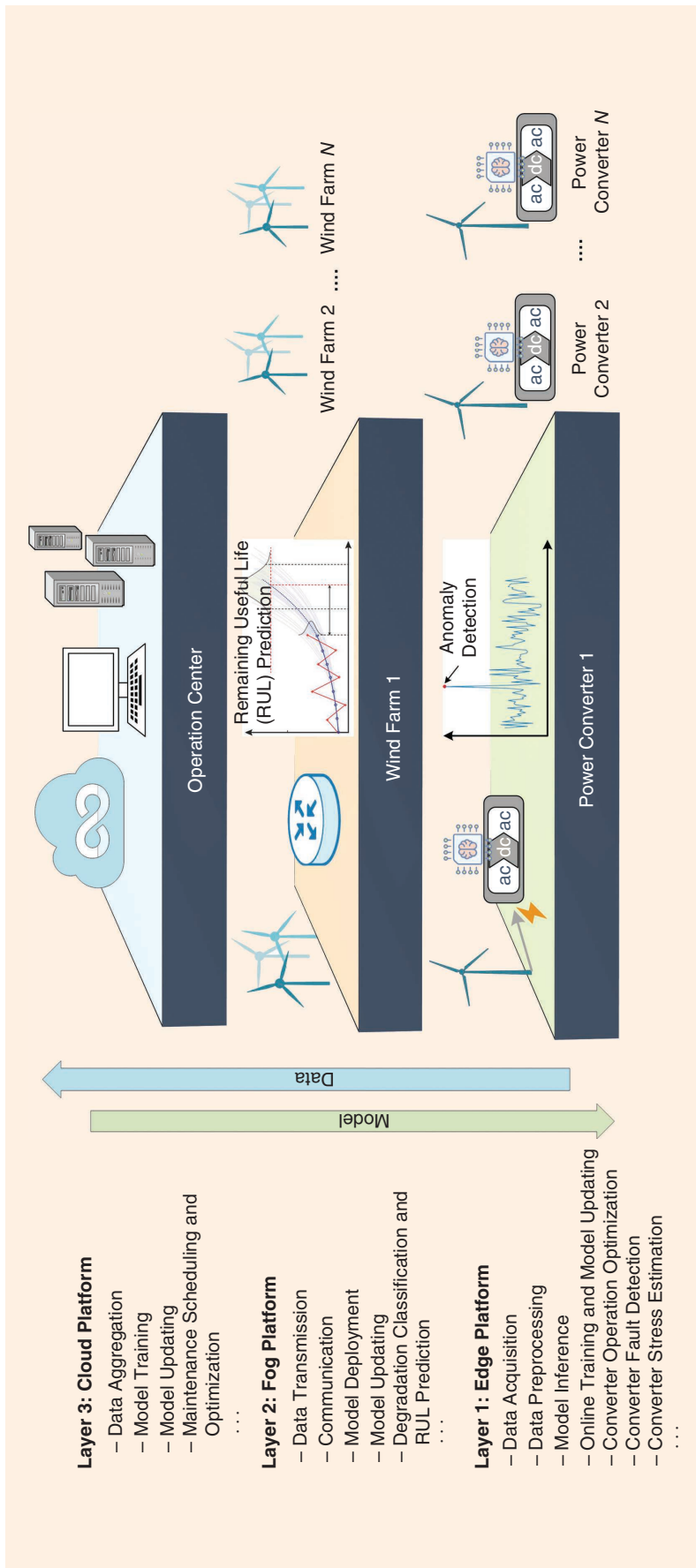


FIG 7 Conceptual three-layer structure for condition monitoring with wind power converters for illustration.

uncertainty quantification of GPR with the 95% confidence interval of output voltage, multiple time-series data of output voltage at normal state can be sampled to make the calculation of extreme values more robust. Generally, it is computationally intensive to calculate the extreme values from multiple time-series. Genetic algorithm, which is an optimization tool, is applied to accelerate searching maximum extreme values from these time series of the output voltage. The model can be trained within 11.657 s, and detection can be completed within microseconds. The accuracy and robustness of the method are verified under settings of injecting noise and various operational temperatures, and all the detection accuracy is above 88.5%.

This case study demonstrates both two steps shown in Figure 2. It investigates different sources of noise factors and variances in environmental conditions. The detection accuracy is quantified and benchmarked with a few other methods, which is a critical performance factor but not often discussed in other case studies.

A Conceptual Framework for Hardware Implementation

Most of the AI tools implementation for condition monitoring generally consists of two phases, i.e., model training and model inference. The model training is usually resource-intensive, which requires a high-performance platform with a powerful computation unit and mass storage. In some cases, e.g., anomaly detection, the model inference is latency-stringent and needs to be performed locally on the PES. Once the condition monitoring data drift from the trained model, the local PES model needs to be re-trained for better performance, i.e., the model updating capability that can dynamically refine itself with the latest data. The implementation can be preferably completed through a hierarchical architecture consisting of three layers, including a cloud platform, fog platform, and edge platform [13].

Figure 7 illustrates a conceptual framework by using wind turbine converters as an example. At the local individual converter-level (i.e., edge

platform), component and converter data can be collected and preprocessed. Real-time decision-making, such as fault detection, protection, operation optimization, and stress estimation, can be performed at this platform. Moreover, models trained offline can be fine-tuned by online training [14] with converter-specific data. The edge platform is locally implemented on the PES with resource-constrained computation units. At the wind farm level (i.e., fog platform), data collected from individual converters can be combined and analyzed for farm-level model training and model updating. Applications such as RUL prediction and degradation classification, which are relatively not time-sensitive, can be performed at this platform. At the operator-level with multiple wind farms (i.e., cloud platform), a massive amount of data are collected for model training. The cloud platform is designed for computation- and data-intensive processing, which is generally equipped with powerful computation units and data warehouse technology. Since it can be remotely accessed and controlled, there is a less geographical requirement on this platform implementation. Maintenance plans requiring complex calculations can be formulated on this layer. It should be noted that the fog platform is complementary to the cloud platform. It is designed as an access point consisting of network gateways and data centers geographically closer to the edge platform. It can coordinate the interoperability and allocate resources between the units on the edge platform, enabling a tightly connected structure. Meanwhile, the fog platform is essential to system resilience in the case of interrupted cloud connection.

For the hardware implementation, the cloud and fog platform solutions can be extended from existing IIoT applications in other fields with matured applications

(e.g., Amazon Web Services, Microsoft Azure IoT). In contrast, the existing implementation of AI tools on the edge platform is limited by far. It requires embedded processors where AI tools can be implemented with sufficient computation power and communication capability. Power consumption, security, flexibility, latency, and cost are the most salient aspects of embedded processors. Figure 8 illustrates several off-the-shelf embeddable AI solutions for the edge platform with feature comparisons. It covers several prevalent embedded solutions, including application-specific integrated circuit (ASIC), field-programmable gate arrays (FPGA), microcontroller unit (MCU), digital signal processor (DSP), graphics processing unit (GPU) and platform combination (e.g., MCU+FPGA). In [15], as an optimization tool, a metaheuristic method real-coded jumping gene genetic algorithm is applied to the parameters of the dynamic model of the photovoltaic panel for data collection. The tool is implemented with a hybrid platform of ARM MCU and FPGA. In [16], a cloud-edge system is implemented for the RUL prediction of MOSFETs in a power converter. An ML tool LSTM is applied for the device degradation modeling. It is implemented with NVIDIA Jetson TX2 for the model inference on the edge layer. The RUL prediction can be completed within 26 ms, and the operational power is less than 2 W.

Outlook on Data-Driven Condition Monitoring for PES

The application of AI for data-driven condition monitoring of PES is increasing in literature. While condition monitoring is demanded in critical power electronic applications, the commercial implementation in power converters is

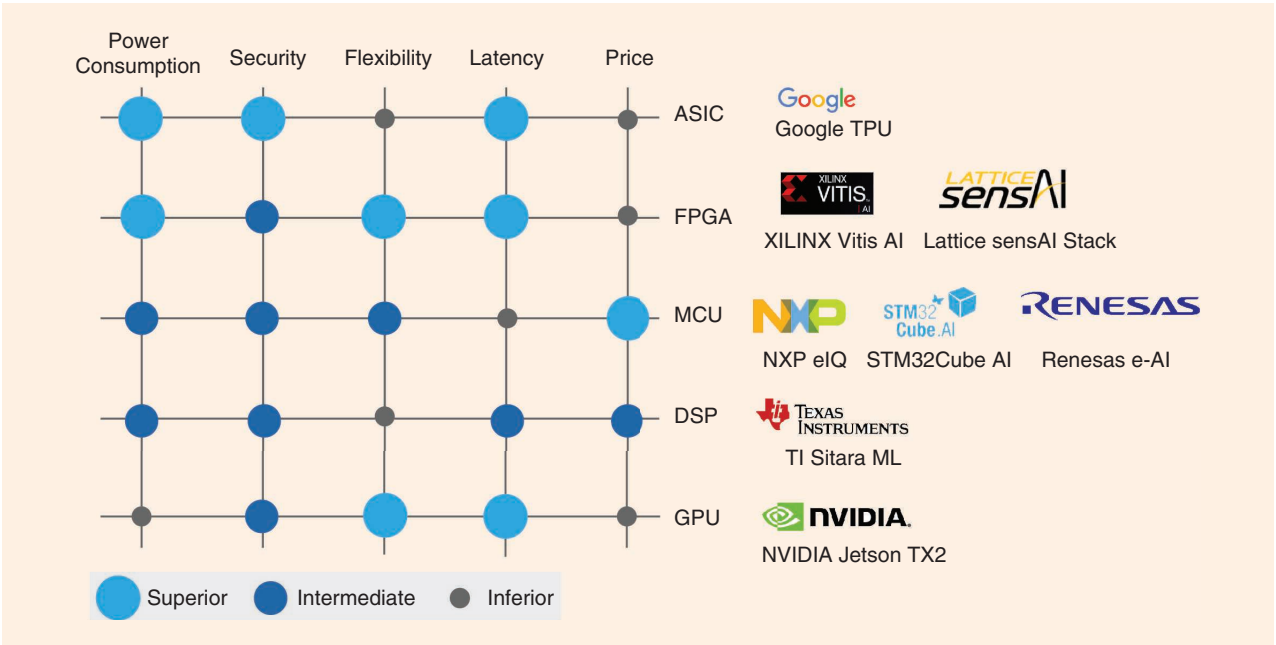


FIG 8 Comparisons of embeddable AI hardware platforms.

still limited. Studies with more comprehensive investigations, considering multiple components, different sources of noises and uncertainties, and quantifying prediction errors are still needed. AI research is currently mainly driven by data-intensive engineering domains, like computer vision and natural language processing. Most of the existing AI solutions need to be specifically tailored for power electronic converter applications. As a result, tiny AI solutions, including tiny data, tiny algorithms, and resource-constrained hardware, are preferable on PES. Several promising directions for future trends are summarized below.

1) Data-Light AI Tools

The generalization and stability of AI tools are highly dependent on the data size. Essentially, the training data require explicitly or implicitly containing the comprehensive system behavior so that AI tools can exploit the underlying knowledge. Currently, most of the research works are carried out based on a relatively small data set. In this case, the applicability of the proposed method in real applications needs to be specifically treated. Although there is an increasing awareness of data collection, the PES is not a data-intensive field to date. As far as we know, there is no well-established data set as the prestigious benchmark data set “*ImageNet*” in image recognition. It thus calls for data-light AI tools that can efficiently be trained with limited data. Transfer learning may help address this challenge since it can efficiently utilize the tiny data from various domains, resulting in a relatively large data set. Meanwhile, research efforts are pressing in academia and industry to formulate a solid data foundation to accelerate the implementation of state-of-the-art deep learning tools in the PES field.

2) Computation-Light AI Tools

Note that the computation unit in PES is with limited computation power. Due to the algorithm complexity, it is challenging to implement state-of-the-art AI tools if the algorithm is applied for time-critical applications. The gap is more evident for legacy systems that are almost impossible to change the design for computation performance improvement. It thus calls for computation-light AI tools that can be implemented on regular computation units with high efficiency. On the other hand, given constrained computation resources, how to accelerate the existing AI tools by using model compression, inference accelerations [17], etc., is crucial.

3) Reliable AI Tools

PES is one of the mission-critical assets in the grid infrastructure. To ensure the reliable implementation of AI tools, the interpretability, accountability, and creditability need to be investigated to mitigate the “black-box” bottleneck. Currently, most of the applications report a high accuracy on a specific task. However, the sensitivity and robustness of the

AI tools are rarely investigated. In addition to the model accuracy, the possible bias and uncertainty need to be assessed quantitatively, and the causality of AI tools needs to be explored. One feasible way is to establish the AI tool within the Bayesian framework to enable the uncertainty quantification capability.

4) Physics-Informed AI Tools

Condition monitoring data are the external representation of PES behavior, which is deterministically governed by physical principles. Another encouraging direction is incorporating well-established physics knowledge and domain experience into the AI tools, suggesting physics-informed AI tools [3]. Merging the available physics knowledge and data for a hybrid method can leverage the benefits from both sides simultaneously. It will potentially benefit the AI tools in the three aspects. Firstly, the incorporation of physics knowledge will offer systematic guidance for collecting data purposefully for the model training, where the data with more insightful information can be focused, whereas the less informative ones will be eliminated. In this way, the data requirements can be alleviated, i.e., data-light AI tools. Secondly, the model structure can be simplified with the incorporation of physics knowledge. The computation-intensive part is possibly replaced with a physical model, suggesting computation-light AI tools. Thirdly, the transparency and explainability of the model can be improved by incorporating physics knowledge to enable reliable AI tools. For this direction, one example is that physical knowledge is added as a regularization term during the training of the neural network, i.e., the physics-informed neural network [3].

5) Distributed Optimization

Research efforts have been devoting to enhance the communication capability between neighboring PES and other computation units in the grid infrastructure. This interconnection will form a multi-agent framework, and the intensive computation can be flexibly distributed on each agent to speed up the task. Such implementation enables distributed optimization. In this way, multiple PES can cooperate to complete an optimization task efficiently. It is beneficial to improve the adaptiveness of the PES since it is highly dependent on optimization efficiency.

Conclusions

This article discusses the efforts made to data-driven condition monitoring of power electronic systems enabled by AI tools. From a data-driven perspective, the main aspects of condition monitoring are presented. The state-of-the-art developments in AI tool applications for condition monitoring and hardware platforms are discussed. Four case studies are given to illustrate the different functions of AI in condition monitoring of PES. An edge-fog-cloud hierarchical architecture is discussed for system implementation. AI brings new possibilities in achieving cost-effective and

robust condition monitoring solutions for PES. The practical applications and field operations are still limited, which are yet to be demonstrated.

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References

[1] S. Yang, D. Xiang, A. Bryant, P. Mawby, L. Ran, and P. Tavner, "Condition monitoring for device reliability in power electronic converters: A review,"

- IEEE Trans. Power Electron.*, vol. 25, no. 11, pp. 2734–2752, Nov. 2010. doi: 10.1109/TPEL.2010.2049377.
- [2] B. K. Bose, "Artificial intelligence techniques in smart grid and renewable energy systems—some example applications," *Proc. IEEE*, vol. 105, no. 11, pp. 2262–2273, Nov. 2017. doi: 10.1109/JPROC.2017.2756596.
- [3] O. Fink, Q. Wang, M. Svensén, P. Dersin, W.-J. Lee, and M. Ducoffe, "Potential, challenges and future directions for deep learning in prognostics and health management applications," *Eng. Appl. Artif. Intell.*, vol. 92, pp. 103678, June 2020. doi: 10.1016/j.engappai.2020.103678.
- [4] S. Zhao, F. Blaabjerg, and H. Wang, "An overview of artificial intelligence applications for power electronics," *IEEE Trans. Power Electron.*, vol. 36, no. 4, 2020. doi: 10.1109/TPEL.2020.3024914.
- [5] Z. Zhao, P. Davari, W. G. Lu, H. Wang, and F. Blaabjerg, "An overview of condition monitoring techniques for capacitors in DC-Link applications," *IEEE Trans. Power Electron.*, vol. 36, no. 4, 2020. doi: 10.1109/TPEL.2020.3023469.
- [6] Y. Peng, S. Zhao, and H. Wang, "A digital twin based estimation method for health indicators of DC-DC converters," *IEEE Trans. Power Electron.*, vol. 36, no. 2, pp. 2105–2118, Feb. 2021. doi: 10.1109/TPEL.2020.3009600.
- [7] Y. Jiang, Y. Yu, and X. Peng, "Online anomaly detection in DC/DC converters by statistical feature estimation using GPR and GA," *IEEE Trans. Power Electron.*, vol. 35, no. 10, pp. 10,945–10,957, Oct. 2020. doi: 10.1109/TPEL.2020.2981500.
- [8] J. Pinto, B. Ozpineci, and R. C. Garcia, "Artificial intelligence applications to power electronics," in *IEEE Energy Convers. Congr. Expo.*, 2019, pp. 1–139.
- [9] M. R. G. Meireles, P. E. M. Almeida, and M. G. Simoes, "A comprehensive review for industrial applicability of artificial neural networks," *IEEE Trans. Ind. Electron.*, vol. 50, no. 3, pp. 585–601, June 2003. doi: 10.1109/TIE.2003.812470.
- [10] S. Zhao, S. Chen, F. Yang, E. Ugur, B. Akin, and H. Wang, "A composite failure precursor for condition monitoring and remaining useful life prediction of discrete power devices," *IEEE Trans. Ind. Inf.*, vol. 17, no. 1, pp. 688–698, Jan. 2021. doi: 10.1109/TII.2020.2991454.
- [11] P. M. Attia et al., "Closed-loop optimization of fast-charging protocols for batteries with machine learning," *Nature*, vol. 578, no. 7795, pp. 397–402, Feb. 2020. doi: 10.1038/s41586-020-1994-5.
- [12] Y. Jiang, Y. Yu, and X. Peng, "Online anomaly detection in DC/DC converters by statistical feature estimation using GPR and GA," *IEEE Trans. Power Electron.*, vol. 35, no. 10, pp. 10,945–10,957, Oct. 2020. doi: 10.1109/TPEL.2020.2981500.
- [13] S. Deng, H. Zhao, W. Fang, J. Yin, S. Dustdar, and A. Y. Zomaya, "Edge intelligence: The confluence of edge computing and artificial intelligence," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7457–7469, Aug. 2020. doi: 10.1109/JIOT.2020.2984887.
- [14] Z. Li, Z. Zheng, and R. Outbib, "A prognostic methodology for power MOSFETs under thermal stress using echo state network and particle filter," *Microelectron. Rel.*, vols. 88–90, pp. 350–354, Sept. 2018. doi: 10.1016/j.microrel.2018.07.137.
- [15] M. Garaj, K. Y. Hong, H. S.-H. Chung, J. Zhou, and A. W.-L. Lo, "Photovoltaic panel health diagnostic system for solar power plants," in *Proc. IEEE Appl. Power Elec. Conf. Expo. (APEC)*, 2018, pp. 1078–1083.
- [16] M. Baharani, M. Biglarbegan, B. Parkhideh, and H. Tabkhi, "Real-time deep learning at the edge for scalable reliability modeling of Si-MOSFET power electronics converters," *IEEE Internet Things J.*, vol. 6, no. 5, pp. 7375–7385, Oct. 2019. doi: 10.1109/JIOT.2019.2896174.